The Bost–Connes System

Zeta as Partition Function

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Abstract

The Bost–Connes (BC) quantum statistical mechanical system is a C*-dynamical system whose partition function is the Riemann zeta function: $Z(\beta) = \zeta(\beta)$ for $\beta > 1$. Its KMS (Kubo–Martin–Schwinger) equilibrium states at inverse temperature β encode the arithmetic of $\mathbb{Q}$: for $\beta > 1$, the unique KMS state recovers the distribution of primes via Dirichlet series; for $\beta = 1$, a phase transition occurs as the system enters a new symmetry class. The zero-temperature limit ($\beta \to \infty$) recovers the Galois action on $\mathbb{Q}^{ab}$. The phase transition at $\beta = 1$ corresponds to the simple pole of ζ at $s = 1$ and the prime number theorem. The nontrivial zeros of ζ appear in the spectral theory of the time evolution, and RH is equivalent to a symmetry property of the KMS states at $\beta = 1/2$. The BC system is a bridge between the analytic approach to RH (zeta function, explicit formula) and the Connes noncommutative geometric approach (adèle class space, absorption spectrum).

1 The C*-Dynamical System

Definition 1 (Bost–Connes algebra [1, 2])**.** The BC algebra $\mathcal{A}$ is the C*-algebra generated by:

- Isometries μ_n for $n \geq 1$: $\mu_n^* \mu_n = 1$, with $\mu_n \mu_n^* = \pi_n$ (projection onto the range).
- Unitaries $e(r)$ for $r \in \mathbb{Q}/\mathbb{Z}$: $e(r)^* = e(-r)$, $e(r)e(s) = e(r+s)$.

The relations:

$$\mu_n e(r) = e(r/n)\mu_n, \qquad \mu_n \mu_m = \mu_{nm}, \qquad \mu_n^* e(r) = \left(\frac{1}{n} \sum_{j=0}^{n-1} e\left(\frac{r+j}{n} \right) \right) \mu_n^*.$$

The time evolution: $\sigma_t(\mu_n) = n^{it}\mu_n$, $\sigma_t(e(r)) = e(r)$.

Remark 2 (Interpretation)*.* The generators μ_n are "multiplication by n" operators on $\mathbb{Q}/\mathbb{Z}$. The generators $e(r)$ are "rotation by r" (characters of $\hat{\mathbb{Z}}$). The time evolution σ_t scales by n^{it}: this is why the Hamiltonian produces $\log n$ in the spectrum. The Hilbert space representation: $\mathcal{H} = \ell^2(\mathbb{N})$, $\mu_n e_k = e_{nk}$, $e(r)e_k = e^{2\pi i r k} e_k$. The Hamiltonian: $He_k = \log k \cdot e_k$.

2 The Partition Function and KMS States

Theorem 3 (Partition function [1]). *In the Hilbert space representation:*

$$Z(\beta) = \mathrm{Tr}(e^{-\beta H}) = \sum_{k=1}^{\infty} k^{-\beta} = \zeta(\beta),$$

convergent for $\beta > 1$. The Riemann zeta function is the partition function of the BC system.

Definition 4 (KMS states). A state φ on $\mathcal{A}$ is a *KMS$_\beta$ state* (at inverse temperature β) if for all $a, b \in \mathcal{A}$:

$$\varphi(a\sigma_{i\beta}(b)) = \varphi(ba).$$

KMS$_\beta$ states are the thermodynamic equilibrium states at temperature $T = 1/\beta$.

Theorem 5 (KMS states of the BC system [1]).

1. *$\beta > 1$: there is a unique KMS$_\beta$ state φ_β:*

$$\varphi_\beta(e(r)) = \frac{1}{\zeta(\beta)} \sum_{n=1}^{\infty} \frac{e^{2\pi i r n}}{n^\beta} = \frac{\mathrm{Li}_\beta(e^{2\pi i r})}{\zeta(\beta)},$$

   where Li_β is the polylogarithm. In particular: $\varphi_\beta(\mu_n^*\mu_n) = 1$ and $\varphi_\beta(\mu_n\mu_n^*) = n^{-\beta}/\zeta(\beta)$. The state "knows" the prime factorization of n through the factor $n^{-\beta}$.

2. *$\beta = 1$: phase transition. The unique KMS$_\beta$ state breaks down; the simplex of KMS$_1$ states is parametrized by probability measures on $\hat{\mathbb{Z}}^*$ (the profinite completion of the integers, invertible elements).*

3. *$0 < \beta < 1$: no KMS states exist.*

4. *$\beta \to \infty$ (zero temperature): the extremal ground states are indexed by the Galois group $\mathrm{Gal}(\mathbb{Q}^{ab}/\mathbb{Q}) \cong \hat{\mathbb{Z}}^*$ (class field theory / Kronecker–Weber).*

Remark 6 (The phase transition at $\beta = 1$). The phase transition at $\beta = 1$ corresponds to: the simple pole of $\zeta(s)$ at $s = 1$; the breakdown of the unique equilibrium state; the onset of spontaneous symmetry breaking of the $\hat{\mathbb{Z}}^*$ symmetry. Physically: at high temperature ($\beta < 1$), no equilibrium; at the critical temperature ($\beta = 1$), the system enters a new phase; at low temperature ($\beta > 1$), unique Gibbs state. The critical temperature $\beta_c = 1$ is the prime number theorem in disguise: the density of primes is $\sim 1/\log n$, and $\sum 1/n$ diverges ($\beta = 1$), while $\sum 1/n^\beta$ converges for $\beta > 1$.

3 The Zeros and the Wall

Theorem 7 (Zeros in the spectral theory [1, 2]). *The nontrivial zeros of ζ appear in the BC system as follows. The operator D on the adèle class space $L^2(C_\mathbb{Q})$ (the Connes scaling operator) has an explicit formula:*

$$\mathrm{Tr}(f(D)) = \int_0^\infty f(u)\, d\mu_{\mathrm{explicit}}(u),$$

where $d\mu_{\text{explicit}}$ is a measure supported on $\{0,1\} \cup \{\operatorname{Im}(\rho)\}$, ρ ranging over nontrivial zeros of ζ. The zeros appear as the support of the spectral measure of D, shifted to the imaginary axis under RH.

Proposition 8 (RH as symmetry of KMS states)**.** *Under RH, the KMS$_\beta$ states satisfy the reflection symmetry: $\varphi_\beta(a^*) = \overline{\varphi_\beta(a)}$ for all $a \in \mathcal{A}$ (this is automatic for any state) and the functional equation $\zeta(\beta) = \chi(\beta)\zeta(1-\beta)$ is implemented by the anti-unitary symmetry $\beta \leftrightarrow 1-\beta$ in the KMS simplex. Without RH (if a zero $\rho = \sigma + i\gamma$ with $\sigma \neq 1/2$ exists), the symmetry of the KMS simplex breaks: the zero contributes an asymmetric pole to the spectral measure of D. RH $\Leftrightarrow$ the KMS simplex at $\beta = 1/2$ is symmetric.*

4 Connection to BN

Theorem 9 (BC and BN reach the same wall)**.** *The BC partition function $Z(\beta) = \zeta(\beta)$ and the BN distance d_N are related:*

$$\log Z(\beta) = -\sum_p \log(1-p^{-\beta}) = \sum_p \sum_{k=1}^{\infty} \frac{p^{-k\beta}}{k} = \int_1^{\infty} x^{-\beta} d(\pi(x)\log x),$$

which, after partial integration, is related to $\int_N^\infty |E(e^u)|^2 e^{-\beta u} du$ for $\beta > 1$. The BN distance $d_N^2 \asymp \int_N^{2N} |E(e^u)|^2/e^{2u}\, du$ is the truncated, normalized version of the contribution of the prime error E to $\log Z(1)$ (the divergent part of the partition function at the critical temperature). BN measures how much of the prime error is learned by step N. BC measures the thermodynamic cost of the whole prime distribution. Both formulations are equivalent to RH.

Remark 10 (Three pictures of the same object). The Riemann zeta function has three faces in this body:

1. *Analytic*: $\zeta(s) = \sum n^{-s}$, zeros at $\rho = 1/2 + i\gamma$ (conjecturally).
2. *Probabilistic/BN*: ζ appears in the Gram matrix symbol $\sigma(t) = |\zeta(1/2+it)|^2/(1/4+t^2)$; the BN distance is the residual after extracting everything arithmetic.
3. *Thermodynamic/BC*: $\zeta(\beta) = Z(\beta)$ is the partition function; the zeros are phase transitions (KMS symmetry breaking points) in the spectral theory.

Each picture makes different aspects visible. All three reach the same wall. The wall is the real part of the zeros.

5 State of the Art

Theorem 11 (What is proved)**.** *For the BC system:*

1. *The partition function is $\zeta(\beta)$ (Theorem 3).*

2. *The KMS$_\beta$ states are completely classified for all $\beta > 0$ (Theorem 5).*

3. *The phase transition at $\beta = 1$ is proved and corresponds to the pole of ζ.*

4. *The zero-temperature states recover the Galois action on $\mathbb{Q}^{ab}$.*

5. *The zeros appear in the spectral measure of the Connes operator (Theorem 7).*

What is not proved:

1. *The KMS symmetry at $\beta = 1/2$ (equivalent to RH).*

2. *The spectral realization of the zeros as genuine eigenvalues of a self-adjoint operator.*

Remark 12 (The wall in thermodynamic language). The BC wall is a symmetry-breaking question. At low temperature ($\beta > 1$): the system has a unique equilibrium state, symmetric under the Galois group. At the critical temperature ($\beta = 1$): symmetry breaks. RH asks: does the symmetry break symmetrically? That is: do the symmetry-breaking modes come in conjugate pairs, reflecting the functional equation of ζ? The answer is yes if and only if the zeros lie on the critical line. We cannot prove it.

References

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